EXPERIMENT 6 – AJAY KARTHICK

AIM

To perform simulation-based investigation of microwave semiconductor devices and frequency-conversion characteristics of a microwave mixer using MATLAB, and to analyze the intermediate frequency, sum frequency, conversion loss, and output spectrum.

OBJECTIVE

1. To develop a simulation model representing the frequency-conversion characteristics of a microwave mixer.
2. To analyze the frequency-dependent behavior of a mixer using radio frequency (RF) and local oscillator (LO) signals.
3. To calculate the difference frequency and sum frequency generated during the mixing process.
4. To calculate conversion loss and determine the intermediate-frequency (IF) output power.
5. To plot the output frequency spectrum and evaluate the frequency-conversion performance of the microwave mixer.

SOFTWARE REQUIRED

MATLAB

PROCEDURE

1. Open MATLAB and create a new script.
2. Define the RF frequency, LO frequency, RF input power, LO power, and IF load impedance.
3. Calculate the intermediate frequency using the difference between the RF and LO frequencies.
4. Calculate the sum frequency generated by the mixing process.
5. Set an assumed conversion loss of 8 dB for the idealized passive-mixer model.
6. Calculate the expected IF output power using the RF input power and assumed conversion loss.
7. Generate a simulated output signal containing the IF and sum-frequency components.
8. Perform a Fast Fourier Transform (FFT) to obtain the output frequency spectrum.
9. Calculate the percentage deviation between theoretical and simulated frequency components.
10. Plot the output spectrum and IF frequency response using MATLAB.
11. Execute the program and observe the numerical results and graphs.

CODE

```
clc;

clear;

close all;

fRF = 10e9;

fLO = 9.5e9;

PRF = -10;

PLO = 7;
```

```
Z0 = 50;
ZIF = 50;
CL_expected = 8;
fIF = abs(fRF - fLO);
fSUM = fRF + fLO;
PIF_expected = PRF - CL_expected;
PSUM = -42;
Fs = 100e9;
T = 200e-9;
N = round(Fs*T);
t = (0:N-1)/Fs;
VIF = sqrt(2*ZIF*10^((PIF_expected-30)/10));
VSUM = sqrt(2*Z0*10^((PSUM-30)/10));
vIF = VIF*cos(2*pi*fIF*t);
vSUM = VSUM*cos(2*pi*fSUM*t);
vOUT = vIF + vSUM;
Y = fft(vOUT);
A = 2*abs(Y(1:N/2+1))/N;
A(1) = A(1)/2;
f = (0:N/2)*Fs/N;
P = A.^2/(2*Z0);
PdBm = 10*log10(max(P,1e-15))+30;
[~,iIF] = min(abs(f-fIF));
[~,iSUM] = min(abs(f-fSUM));
fIF_sim = f(iIF);
fSUM_sim = f(iSUM);
PIF_sim = PdBm(iIF);
CL_sim = PRF-PIF_sim;
devIF = abs(fIF_sim-fIF)/fIF*100;
devSUM = abs(fSUM_sim-fSUM)/fSUM*100;
subplot(2,1,1);
plot(f/1e9,PdBm,'b','LineWidth',1.5);
hold on;
plot(fIF_sim/1e9,PIF_sim,'ro','MarkerFaceColor','r');
```

OUTPUT



REFLECTION

This experiment helped me understand the frequency-conversion characteristics of a microwave mixer and the operating principles of RF and LO signals. I learned how to calculate the intermediate frequency and sum frequency generated during mixing and determine the conversion loss and IF output power. The MATLAB simulation helped me visualize the output frequency spectrum and compare theoretical and simulated frequency components. This experiment also improved my understanding of the importance of RF power, LO power, and impedance matching in microwave frequency-conversion applications.

RESULT

The simulation-based investigation of microwave mixer frequency conversion was performed using MATLAB. The intermediate frequency of 0.5 GHz and sum frequency of 19.5 GHz were calculated and represented in the simulated output spectrum. Using an assumed conversion loss of 8 dB, the expected IF output power was calculated as −18 dBm. The frequency-conversion characteristics were visualized using MATLAB plots.

Experiment-6

$f_{RF} = 10\,GHz$
$f_{LO} = 9.5\,GHz$
$P_{RF} = -10\,dBm$
$Z_{IF} = 50\,\Omega$

i) Intermediate Frequency

$$f_{IF} = |f_{RF} - f_{LO}|$$

$$\Rightarrow 10 - 9.5$$

$$\Rightarrow 0.5_{IF}$$

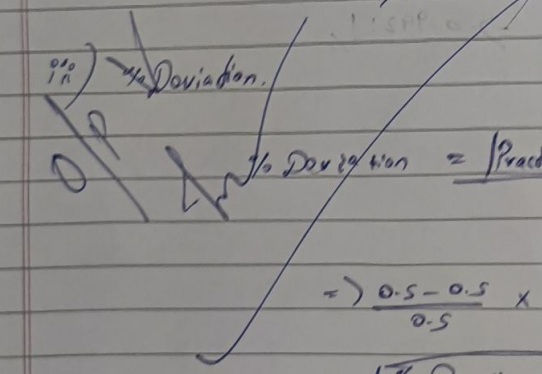
ii) Sum frequency

$$f_{sum} = f_{RF} + f_{LO}$$

$$\Rightarrow 10 + 9.5$$

$$\Rightarrow 19.5\,HZ$$

iii) % Deviation.



$$\% \text{ Deviation} = \frac{|\text{Practical} - \text{Theoretical}|}{|\text{Theoretical}|} \times 100$$

$$\Rightarrow \frac{0.5 - 0.5}{0.5} \times 100$$

$$\boxed{\% \text{ Deviation} = 0}$$

iv) Conversion loss

$$C_L = P_{RF,\,in} - P_{IF,\,out}$$

$$P_{RF,\,in} = -10\,dB$$

$$\boxed{P_{IF} = -18\,dBm}$$